

PROJECT REPORT

Analysis of Student Academic Performance Using R and RStudio

A Beginner-Level Data Analysis and Data Visualization Project

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1. Abstract

This report presents an exploratory data analysis (EDA) and visualization project conducted in R and RStudio, using a dataset of 200 students that records study habits, attendance, sleep, family background, and academic outcomes. The objective is to demonstrate, at a beginner level, the complete workflow of a data analysis project: importing data, cleaning and exploring it, computing summary statistics, and creating visualizations that reveal patterns in the data. The analysis finds a clear positive relationship between weekly study hours and final exam score, a smaller positive relationship with attendance, and a lower average score among students who hold a part-time job. These findings are consistent with common-sense expectations about academic performance and serve to illustrate how simple statistical and visual techniques can validate real-world intuitions using data.

2. Introduction

Data analysis is the process of inspecting, cleaning, and modeling data to discover useful information and support conclusions. For students and early researchers, R is one of the most widely used tools for this purpose because of its strong statistical foundations and the ggplot2 library, which is considered one of the most flexible and widely adopted data visualization frameworks.

This project was designed as an introductory, hands-on exercise: it uses a single, moderately sized dataset that is easy to understand (student academic performance) so that a first-time user of R can focus on learning the tools rather than on understanding a complex domain.

3. Objectives

- To import and inspect a real-world-style dataset in RStudio.
- To compute descriptive statistics (mean, median, correlation) for key variables.
- To visualize the distribution of a numeric variable using a histogram.
- To compare categorical groups using bar charts and boxplots.
- To examine the relationship between two numeric variables using a scatter plot and trend line.
- To draw simple, evidence-based conclusions about factors associated with academic performance.

4. Dataset Description

The dataset, `student_performance.csv`, contains 200 rows (one per student) and 11 columns. It was constructed to reflect commonly studied factors in academic performance research, with built-in realistic relationships between study habits and outcomes.



The dataset contains no missing values, making it suitable for a first project without requiring advanced data-cleaning techniques.

5. Methodology

The analysis followed a standard exploratory data analysis workflow, carried out entirely in RStudio using the tidyverse collection of packages:

- Data Import: The CSV file was read into R using `read_csv()` from the `readr` package.
- Data Inspection: Structure and summary statistics were examined using `str()` and `summary()` to understand variable types and ranges.
- Data Quality Check: `colSums(is.na(...))` confirmed the absence of missing values.
- Descriptive Statistics: Mean, median, and Pearson correlation coefficients were computed for key numeric variables.
- Group Comparison: `dplyr`'s `group_by()` and `summarise()` functions were used to compare average scores across categorical groups (e.g., Gender, PartTimeJob).
- Visualization: `ggplot2` was used to construct a histogram, a bar chart, a boxplot, and a scatter plot with a linear trend line.

6. Analysis and Results

6.1 Overall Score Distribution

The `FinalExamScore` variable ranges from approximately 20 to 100 across the 200 students, with a mean of roughly 65 and a roughly bell-shaped (approximately normal) distribution when plotted as a histogram. A smaller cluster of students score below 45, which corresponds to the Fail category.

6.2 Pass vs Fail

Out of 200 students, 182 (91%) passed and 18 (9%) failed. This class imbalance is realistic (most students in a typical class do pass) and is an important observation to note before drawing conclusions from group comparisons, since the Fail group is smaller and its average is based on fewer data points.

6.3 Study Hours and Exam Score

A scatter plot of `StudyHoursPerWeek` against `FinalExamScore`, with a linear trend line overlaid, shows a clear upward-sloping relationship: students who report studying more hours per week tend to score higher on the final exam. The Pearson correlation coefficient between these two variables is approximately +0.42, indicating a positive, moderate-strength relationship — meaningful, but not strong enough on its own to fully determine the outcome, meaning other factors also play a role.

6.4 Attendance and Previous GPA

`AttendancePercent` shows a weaker positive correlation with final exam score (approximately +0.23), and `PreviousGPA` shows a similar positive correlation (approximately +0.26). Both suggest that consistent attendance and prior academic performance are associated with better outcomes, but neither is as strong a predictor on its own as weekly study hours.

6.5 Part-Time Job and Exam Score

The boxplot comparing `FinalExamScore` between students with and without a part-time job shows that students without a part-time job tend to have a higher median score and a tighter interquartile range, while students with a part-time job show a lower median and a wider spread. This is consistent with the intuition that outside work commitments can reduce the time and energy available for academic work, and matches how the dataset's `PartTimeJob` effect was defined.

6.6 Sleep Hours

Sleep hours show almost no linear correlation with final exam score in this dataset (approximately -0.08). This is a useful negative finding: it demonstrates that not every variable in a dataset will show a strong relationship with the outcome of interest, and that correlation analysis is as valuable for ruling out weak relationships as it is for confirming strong ones.

6.7 Summary of Correlations with Final Exam Score

7. Visualizations Produced

The following four core visualizations were generated as part of this project (see `analysis.R` and the accompanying instructional document for full code):

8. Discussion

The results of this project align with common expectations in education research: time invested in studying is the strongest observed predictor of exam performance among the variables tested, attendance contributes a smaller positive effect, and competing time commitments such as a part-time job are associated with modestly lower performance. Sleep, while important for wellbeing generally, did not show a strong linear relationship with exam score in this sample.

It is important to note the limitations of this analysis. The dataset is moderate in size (200 students) and was used here primarily as a learning tool; correlation does not imply causation, and unmeasured factors (such as subject difficulty, individual study techniques, or test anxiety) could influence the observed relationships. Nonetheless, the project successfully demonstrates how a structured, step-by-step data analysis workflow in R can surface interpretable, actionable patterns from raw data.

9. Conclusion

This project demonstrated a complete beginner-level data analysis and visualization workflow in RStudio: importing a dataset, exploring its structure, computing descriptive statistics, and building four distinct types of visualizations using `ggplot2`. The analysis found that study hours and, to a lesser extent, attendance are positively associated with exam performance, while holding a part-time job is associated with a modest decrease in performance. These findings show how even simple, well-chosen visualizations and summary statistics can turn raw tabular data into clear, communicable insights — the foundational skill this project set out to teach.

10. Tools and References

- R Core Team — The R Project for Statistical Computing (<https://www.r-project.org>)
- Posit (RStudio) — RStudio Desktop IDE (<https://posit.co>)
- Wickham, H. — `ggplot2`: Elegant Graphics for Data Analysis
- Wickham, H. et al. — `dplyr` and the tidyverse collection of R packages
- Dataset: `student_performance.csv` — synthetically generated for this beginner project